I want to solve the economic dispatch for the Spanish electricity system and do an analysis on the results:

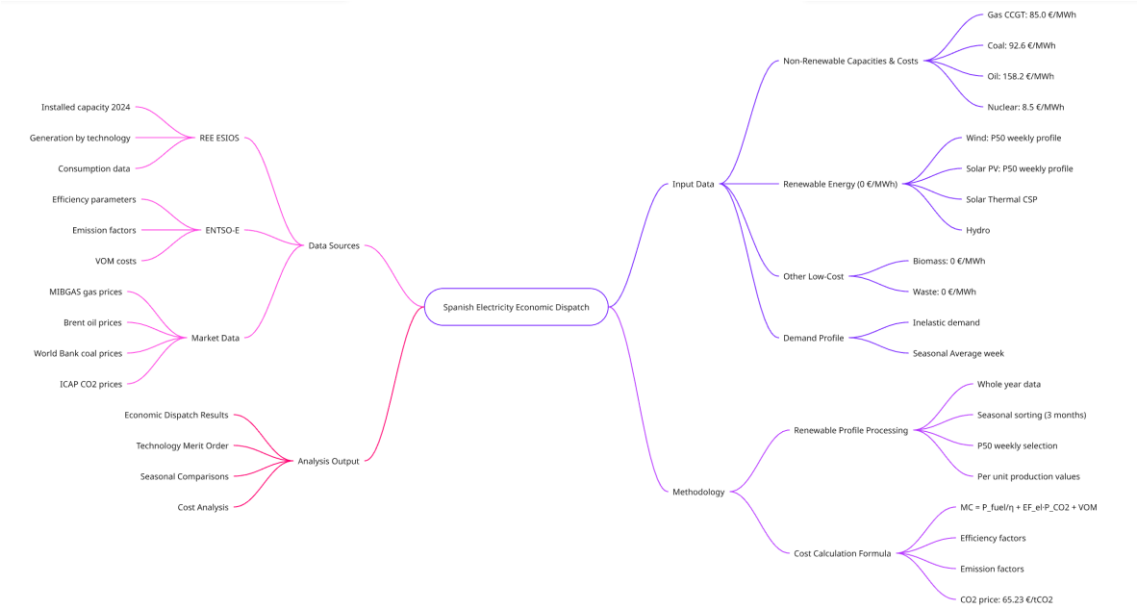
1.- I need the capacities of the Non-renewable sources and their operation cost as input data.

2.-For the renewable energy (wind ans solar) I will do the following:

We will take a whole year profile for both technologies, then we´ll get the P50 weekly production of a season: So, if a season has 3 months, we will sort all weeks in it and take the P50.

All the production values will be in per unit and the capacity installed will be given as an input. (we will assume the renewable cost as 0)

3.-We will assume an inelastic demand also given as an input that will depend on the season and will be chosen as well as the renewable sources (the P50 week of a season)



Solutions:

Installed capacity 2024:
https://www.esios.ree.es/en/analysis/10300?vis=3&start_date=01-03-2024T00%3A00&end_date=31-03-2024T23%3A55&compare_start_date=04-01-2024T00%3A00&groupby=month&compare_end_date=03-03-2024T22%3A55

Generation (by technology) and consumption 2024:
https://www.esios.ree.es/en/analysis/10024?compare_indicators=369%2C370%2C92%2C91%2C79%2C95%2C88%2C90%2C96%2C83%2C82%2C80%2C81%2C87%2C97%2C71%2C2131%2C72%2C77%2C75%2C78%2C76%2C74%2C86%2C93%2C94%2C84%2C85%2C89%2C73%2C2143%2C2154&start_date=01-01-2024T00%3A00&geoids=&vis=1&end_date=31-12-

2024T23%3A55&compare_start_date=31-12-2023T00%3A00&groupby=hour

Variable cost definition (Fuel + CO2+ O&M) (entso-e):

https://consultations.entsoe.eu/entso-e-general/proposal-for-european-resource-adequacy-assessment/supporting_documents/191205_Methodology%20for%20European%20Resource%20Adequacy%20Assessment_public%20consultation.pdf

Common structure (thermal units)

For thermal technologies (gas/oil/coal/biomass/waste if modelled as thermal):

$$MC = \frac{P_{fuel}}{\eta} + EF_{el} \cdot P_{CO2} + VOM$$

- P_{fuel}= fuel price in **€/MWh_th**
- η= electrical efficiency (NCV basis)
- EF_{el}= emissions in **tCO_2/MWh_e**
- P_{CO2}= **€/tCO_2**
- VOM= variable O&M in **€/MWh_e**

And when an emission factor is given per fuel energy (ENTSO-E gives **kg/GJ_fuel**), convert to electricity basis:

$$EF_{el}\ [t/MWh_e] = \frac{EF_{fuel}\ [kg/GJ] \cdot 3.6}{\eta \cdot 1000}$$

ENTSO-E provides η, $EF_{fuel}$and VOM by technology:

https://www.entsoe.eu/Documents/TYNDP%20documents/TYNDP2018/Scenarios%20Data%20Sets/Input%20Data.xlsx

CO2 price:

https://icapcarbonaction.com/en/ets/eu-emissions-trading-system-eu-ets

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1) Gas (CCGT)

Equation

$$MC_{gas} = \frac{P_{gas}}{\eta_{gas}} + EF_{gas,el} \cdot P_{CO2} + VOM_{gas}$$

Parameters (values used)

- $\eta_{gas} = 0.58$(ENTSO-E "Gas – CCGT new").
- $EF_{fuel} = 57$kgCO$_2$/GJ (ENTSO-E gas).
- $VOM_{gas} = 1.6$€/MWh (ENTSO-E "Gas – CCGT new").
- $P_{gas} \approx 35$€/MWh_th (REE states "MIBGAS price for 2024 was around €35/MWh").

https://www.sistemaelectrico-ree.es/en/spanish-electricity-system/markets/day-ahead-market

- $P_{CO2} = 65.23$€/tCO$_2$ (ICAP 2024 average secondary market price).

Computed (example)

- $MC_{gas} \approx 85.0$€/MWh_e

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2) Oil (modeled as Heavy Fuel Oil steam unit)

Equation

$$MC_{oil} = \frac{P_{oil}}{\eta_{oil}} + EF_{oil,el} \cdot P_{CO2} + VOM_{oil}$$

Parameters (values used)

- $\eta_{oil} = 0.40$(ENTSO-E "Heavy oil – old 2").
- $EF_{fuel} = 78$kgCO$_2$/GJ (ENTSO-E oil).
- $VOM_{oil} = 3.3$€/MWh (ENTSO-E heavy oil).
- Brent 2024 average $= 78.73$\$/bbl (REE).

  https://www.sistemaelectrico-ree.es/en/spanish-electricity-system/markets/day-ahead-market

- USD/EUR 2024 annual avg $= 1.0824$USD per EUR (Bundesbank, based on ECB reference rates).
- Energy per barrel: EIA gives **5,689,000 Btu per barrel of crude**.
- Conversion convention: 1 MMBtu $= 0.293071$ MWh(Platts methodology note).
- $P_{CO2} = 65.23$€/tCO$_2$.

(Those give an implied P_{oil}≈ 43.6 €/MWh_th for this example.)

Computed (example)

- $MC_{oil} \approx 158.2$€/MWh_e

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3) Coal (Soft Coal / Lignite)

Equation

$$MC_{coal} = \frac{P_{coal}}{\eta_{coal}} + EF_{coal,el} \cdot P_{CO2} + VOM_{coal}$$

Parameters (values used)

- $\eta_{coal} = 0.40$
  (Typical efficiency for lignite / soft coal plants)
  Source: ENTSO-E TYNDP Input Data
  https://www.entsoe.eu/Documents/TYNDP%20documents/TYNDP2018/Scenarios%20Data%20Sets/Input%20Data.xlsx

- $EF_{fuel} = 101$kgCO_2/GJ
  (Emission factor for lignite / soft coal)
  Source: ENTSO-E TYNDP Input Data
  https://www.entsoe.eu/Documents/TYNDP%20documents/TYNDP2018/Scenarios%20Data%20Sets/Input%20Data.xlsx

- $VOM_{coal} = 3.3$€/MWh
  (Typical variable O&M for coal plants)
  Source: ENTSO-E TYNDP Input Data
  https://www.entsoe.eu/Documents/TYNDP%20documents/TYNDP2018/Scenarios%20Data%20Sets/Input%20Data.xlsx

- $P_{coal} \approx 12$ €/MWh_{th}
  (Typical lignite/soft coal fuel price equivalent)
  Source: World Bank Commodity Markets "Pink Sheet"
  https://thedocs.worldbank.org/en/doc/18675f1d1639c7a34d463f59263ba0a2-0050012025/related/CMO-Pink-Sheet-January-2026.pdf

- $P_{CO2} = 65.23$€/tCO_2
  (EU ETS 2024 average price)
  Source: ICAP EU ETS
  https://icapcarbonaction.com/en/ets/eu-emissions-trading-system-eu-ets

  $MC_{coal} \approx 92{,}6$ €/MWh_e

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4) Hydro

In pure economic dispatch models, hydro is often treated as **very low marginal cost** (no fuel; small variable O&M), but **the real driver is water opportunity cost**, which depends on reservoir constraints and is not a fixed parameter.

Simple dispatch approximation

$$MC_{hydro} \approx 0 (\text{or a small } VOM)$$

Value used (example)

- $MC_{hydro} = 0$€/MWh_e (typical assumption: very low variable production cost).

https://economy-finance.ec.europa.eu/system/files/2023-06/dp187_en_energy%20markets.pdf (Used also in all renewables)

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5) Bio

Bioenergy plants are thermal units, so you *can* use the thermal equation, but **fuel cost varies hugely** by feedstock and contract, and ETS treatment depends on biomass definition (often zero-rated under ETS rules when qualifying as "biomass").

If you don't have a defensible biomass fuel-price series yet, a common simplified dispatch assumption is "near-zero marginal cost" (or fixed adder).

Simple placeholder (example)

$$MC_{bio} \approx 0$$

Citing low variable cost assumption for renewables/low marginal-cost generation (as a simplification until you source biomass fuel costs).

Value used

- $MC_{bio} = 0$€/MWh_e

(If you want, we can do a proper bio model next: $P_{bio}$from pellet/biomass indices + η+ VOM.)

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6) Waste

Waste-to-energy marginal cost is also site-specific (gate fees can make "fuel price" effectively negative; ETS treatment depends on fossil fraction and scope). If you don't model that market explicitly, it's commonly approximated as low/near-zero marginal cost in simplified dispatch.

Placeholder (example)

$$MC_{waste} \approx 0$$

(using the same "very low variable production cost" simplification family).

Value used

- $MC_{waste} = 0$€/MWh_e

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7) Wind

Wind has **near-zero marginal cost** (no fuel, very low variable cost), and typically bids at/near that level.

$$MC_{wind} \approx 0$$

Value used

- $MC_{wind} = 0$€/MWh_e

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8) Solar PV

Same logic: near-zero marginal cost; typically bid near that level (including in Iberian markets).

$$MC_{PV} \approx 0$$

Value used

- $MC_{PV} = 0$€/MWh_e

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9) Solar thermal (CSP)

CSP has no fuel (unless hybridized with gas); variable cost is usually treated as very low in simplified dispatch.

$$MC_{CSP} \approx 0$$

Value used

- $MC_{CSP} = 0$€/MWh_e (same "low variable production cost" rationale).

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10) Nuclear

Equation

$$MC_{nuclear} = F_{nuclear} + VOM_{nuclear}$$

(No fuel/efficiency term in dispatch formulation since nuclear fuel cycle cost is already expressed per MWh_e. No CO_2 cost.)

Parameters (values used)

- $F_{nuclear} = 6.5$€/MWh
  (Typical fuel cycle cost range 5–8 €/MWh)
  Source: IEA/NEA *Projected Costs of Generating Electricity 2020*
  https://www.iea.org/reports/projected-costs-of-generating-electricity-2020

- $VOM_{nuclear} = 2$€/MWh
  Source: ENTSO-E TYNDP Input Data (Nuclear archetype VOM)
  https://www.entsoe.eu/Documents/TYNDP%20documents/TYNDP2018/Scenarios%20Data%20Sets/Input%20Data.xlsx

- $EF_{nuclear} = 0$
  (No combustion emissions under EU ETS)
  Source: EU ETS framework
  https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets_en

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Summary of computed marginal costs (example, 2024-style inputs)

- **Gas (CCGT new): ~85.0 €/MWh**
- **Coal (Soft coal new): ~92,6 €/MWh**
- **Oil (Heavy oil old2): ~158.2 €/MWh**
- **Hydro/Bio/Waste/Wind/PV/CSP: 0 €/MWh** (simplified assumption)
- **Nuclear: 8,5 €/MWh**